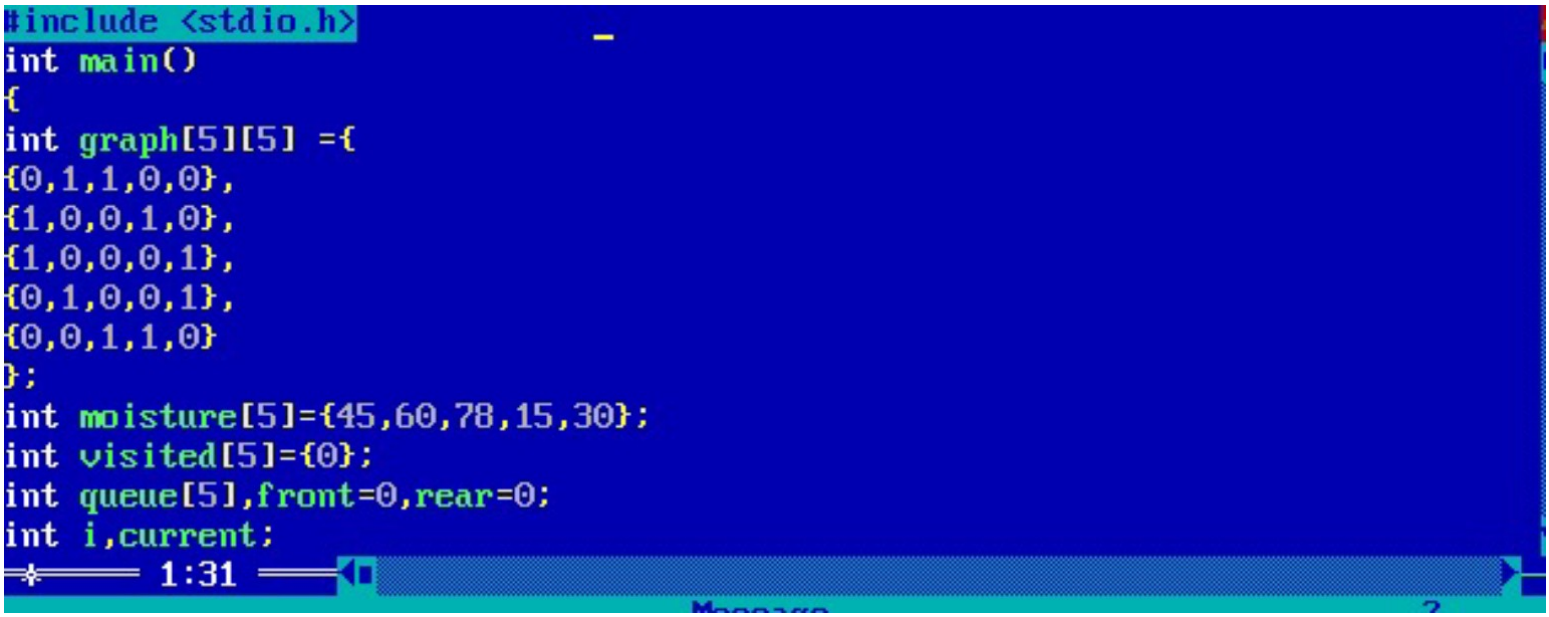
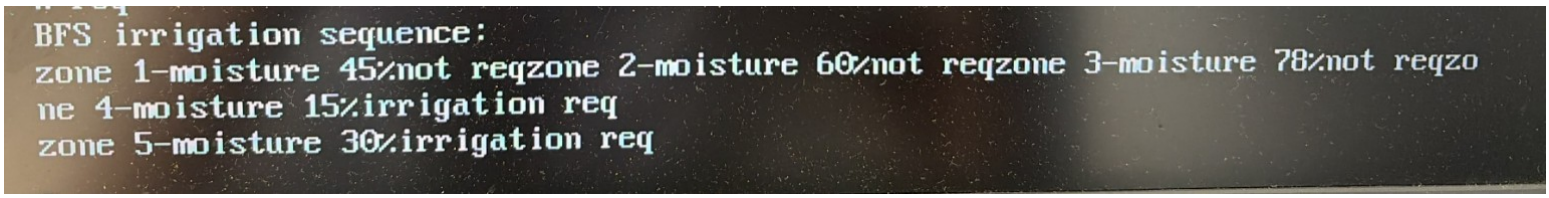

#include <stdio.h>
int main()
{
int graph[5][5] ={
{0,1,1,0,0},
{1,0,0,1,0},
{1,0,0,0,1},
{0,1,0,0,1},
{0,0,1,1,0}
};
int moisture[5]={45,60,78,15,30};
int visited[5]={0};
int queue[5],front=0,rear=0;
int i,current;
1:31

```
queue[rear++]=0;
visited[0]=1;
printf("BFS irrigation sequence:\n");
while(front<rear)
{
current =queue[front++];
printf("zone %d-moisture %d%%",current +1, moisture[current]);
if (moisture[current]<40)
printf("irrigation req \n");
else
printf("not req");
for(i=0;i<5;i++)
{
if(graph[current][i]==1&& visited[i]==0)
```

28:31

Message 2

```
printf("irrigation req \n");
else
printf("not req");
for(i=0;i<5;i++)
{
if(graph[current][i]==1&& visited[i]==0)
{
visited[i]=1;
queue[rear++]=i;
}
}
}
return 0;
}
```


BFS irrigation sequence:
zone 1-moisture 45%not reqzone 2-moisture 60%not reqzone 3-moisture 78%not reqzo
ne 4-moisture 15%irrigation req
zone 5-moisture 30%irrigation req